

Hiero Workflow App

Architecture-First Strategy & Migration Plan

LFDT Mentorship 2026 · June 15 – November 30, 2026

LFDT Mentorship 2026

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Document Purpose

This document presents the corrected product architecture, phased delivery strategy, community issue rollout plan, and six-month LFDT mentorship execution plan for the Hiero Workflow App. It supersedes the earlier migration-first document.

Core correction: build a small, robust V2 GitHub App from the start. Do not start with a V1.5 workflow migration wrapper and hope it becomes an App later.

What Was Wrong Before

The earlier document was still a migration/testing-oriented document rather than an architecture-first document. The core problems were:

- Dry-run, canary, and timeline sections appeared *before* the architecture.
- Python review-sync was the first product story, not **/assign**.
- GitHub Actions were framed as the product architecture rather than a testing adapter.
- The architecture diagram was missing the listener → router → dispatcher pipeline.
- Components were coupled; workflows were entangled with policy logic.

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1. Executive Summary

1.1. The Core Insight

Sophie's feedback was precise: the document should open with the final product architecture, not with a migration plan. Speed is not the goal. The correct approach is to build a small V2 application with the right architectural skeleton, then grow capability one module at a time.

Corrected Strategy In One Sentence

Build a GitHub App with a clean event pipeline (listener → normalizer → router → dispatcher → policy module → executor → audit), start with **/assign** as the first product slice, and add PR quality and lifecycle modules only after the app shell is proven.

1.2. Old vs. Corrected Framing

Old framing (to replace)	Corrected framing
Shared GitHub Action / canary is the product centre.	GitHub App architecture is the product centre. Actions and canaries are compatibility/testing paths only.
Python review-sync first because the canary worked.	/assign first: smaller product slice, explicitly identified by Sophie as the first candidate.
Migration timeline appears before architecture.	Architecture comes first; testing, canary, and migration timeline appear after the product model.
Move existing services into a shared repo quickly.	Build the listener → router → dispatcher path first, then attach one policy module at a time.
Config is mainly a file for extracted constants.	Config is the app policy surface: feature flags, thresholds, doc links, labels, and module settings validated by schema.

Dry-run and canary as architectural choices.

Dry-run and canary are validation techniques. They belong in the testing section, not the architecture section.

1.3. Six-Month Delivery Summary

Phase	Focus	Key Deliverable	Timeline
0	Architecture agreement	Skeleton agreed; App registered; hosting selected	Pre Jun 15
1	App shell	Listener → audit pipeline; CI/CD deploy	Jun 15 – Jul 15
2	/assign minimal	Working /assign in sandbox	Jul 16 – Aug 10
3	/assign guards	Full guard coverage, parity fixtures	Aug 11 – Aug 31
4	PR quality module	DCO, GPG, title, linked-issue checks	Sep 1 – Oct 15
5	Lifecycle modules	Issue/PR lifecycle (2 modules)	Oct 16 – Nov 14
6	Wider rollout	Per-repo adoption, community issues	Nov 15 – Nov 30

1.4. Answering The Core Strategic Questions

Sophie posed four questions that sit above technical implementation. They test whether the candidate can think about the *system*, not just the code. This subsection answers each one directly.

Question	Answer
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How do you migrate all the code?

We do not migrate all the code at once. We migrate *reusable policy decisions* one vertical feature slice at a time. The first slice is **/assign**: we build the full app path (listener → normalizer → router → dispatcher → policy module → executor → audit) for that single command before touching any other automation. Each subsequent module (PR quality, issue lifecycle, PR lifecycle, progression, review) follows the same pattern: prove in sandbox, establish fixture-parity with existing Python and C++ behaviour, dry-run pilot, then controlled production enablement. CI/CD workflows, runner configuration, checkout, build scripts, and repository secrets are *never* migrated — they remain in the SDK repositories permanently (see Section 6 for the full boundary table).

What is reusable?

The entire App Shell is reusable across every Hiero repository: webhook listener, event normalizer, router, dispatcher, config engine, executor, and audit logger. These eight components are built once, deployed once, and shared by all installed repositories. Policy modules are also reusable — each module is a pure decision function parameterised by per-repo config. For example, the Assignment Policy module serves Python, C++, and any future SDK with the same code, but with different thresholds, labels, and team names loaded from each repository's `.github/hiero-automation.yml`. The config schema, the audit log format, the idempotency and retry logic in the executor, and the security verification in the listener are all shared infrastructure that never needs per-repo duplication.

What needs to be torn up?

Three categories of legacy code must be retired: **(1)** per-repo automation scripts (e.g. `assign.py`, `review_sync.py`, and their duplicates across Python and C++ SDKs), **(2)** GitHub Action workflow wrappers that embed policy logic inside YAML steps instead of delegating to a centralised module, and **(3)** scattered API retry and error-handling code that lacks idempotency guarantees and bot-marker checks. None of this code is deleted on day one — each piece is retired only *after* its replacement module passes fixture-parity tests and a controlled pilot. Section 6.4 provides the full teardown inventory with phase-linked retirement dates.

How do you make the solution future-looking?

Five architectural choices ensure long-term durability: **(1)** policy modules are pure functions with no side effects and no cross-module coupling, making them trivially replaceable or extensible; **(2)** configuration is versioned via JSON schema with a 30-day backward-compatibility window, so schema evolution does not break installed repositories; **(3)** the executor centralises all GitHub API writes behind pinned API versions (`X-GitHub-API-Version`) and idempotency keys, insulating every module from upstream API drift; **(4)** the audit logger provides a complete decision trail for regression analysis, debugging, and maintainer trust across releases; **(5)** the module registry is open — adding a new policy module requires only implementing the module interface and registering a route, with zero changes to the listener, router, or dispatcher.

2. Product Goal And Non-Goals

2.1. Goal

Create a maintainable Hiero Workflow App that handles common maintainer workflows across Python, C++, and later other Hiero repositories *without* copying scripts into every SDK. The app must be modular enough to start with `/assign` and later support PR quality, issue cleanup, PR cleanup, progression, and review workflows, all from a single installable GitHub App with a shared event pipeline.

2.2. Non-Goals

- Do not migrate every Python and C++ workflow at the start.
- Do not hide repository security controls or ownership inside a wrapper.
- Do not make dry-run/canary the architecture — they are validation techniques.
- Do not build a giant app with every feature before the first robust slice works.
- Do not make configuration an arbitrary scripting surface.
- Do not make the mentorship output a collection of workflow migrations; make it a small, trustworthy product platform.

2.3. Success Criteria For The Mentorship

Metric	Target	Evidence
App shell stability	Synthetic webhook events route correctly through all eight pipeline stages	Passing integration tests and audit log output
<code>/assign</code> parity	Python and C++ expected assignment outcomes match app behaviour	Fixture-backed parity test suite

Maintenance friction	At least one shared policy can be updated centrally instead of per-repo patching	Central change plus per-repo adoption evidence
Pilot safety	No contributor-facing disruption during /assign sandbox pilot	Pilot logs, rollback drill, maintainer sign-off
Testing discipline	90%+ coverage for all state-mutating logic and policy decisions	Coverage reports and CI gates
Community leverage	Structured issue backlog exists for contributors with clear safe-to-open phases	Issue map, labels, and phase-linked backlog

3. End-State Architecture

3.1. Architecture Overview

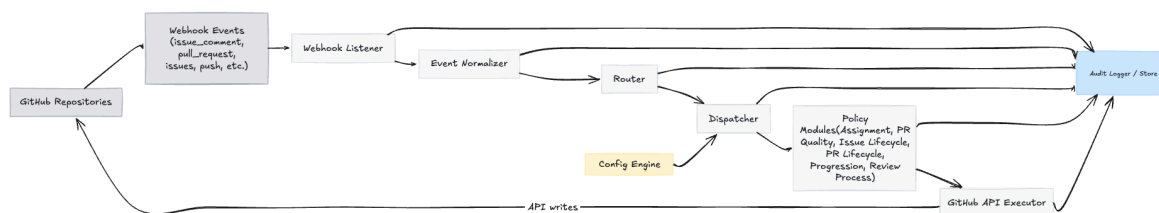


Figure 1: End-State App Architecture

3.2. Primary Components

Component	Responsibility
Webhook Listener	Receives GitHub webhook events over HTTPS. Verifies HMAC-SHA256 signature, installation ID, repository membership, event type, delivery ID, and basic payload shape. This is the first and primary trust boundary. Rejects malformed or unauthorised payloads immediately, before any routing logic runs.
Event Normalizer	Converts raw GitHub webhook payloads into a stable internal event model (NormalizedEvent). Downstream policy modules never depend directly on raw webhook shape, so GitHub API changes require only normalizer updates.

Router	Maps normalized events and parsed commands (e.g. <code>/assign</code> , <code>/unassign</code> , <code>/blocked</code>) to named product routes. Routes are registered declaratively; adding a new command does not require changes to the listener or dispatcher.
Dispatcher	Loads validated config for the target repository, selects the correct policy module from a module registry, calls it with the normalized event context, and forwards approved operations to the executor. Handles module-not-found, config-invalid, and policy-error paths without crashing.
Config Engine	Loads <code>.github/hiero-automation.yml</code> from the target repository via the GitHub API. Validates against a versioned Zod JSON schema, generating TypeScript types. Applies safe conservative defaults. Exposes only typed, validated settings to modules. Rejects unknown high-risk fields; warns on unknown low-risk fields.
Policy Modules	Small, independent modules with no cross-module coupling. Each module is a pure decision function: it receives a validated event context and config, and returns a list of approved operations (<code>ApprovedOperation</code>). Modules do not call GitHub directly. Current planned modules: Assignment, PR Quality, Issue Lifecycle, PR Lifecycle, Progression, Review Process. Each module exports a static manifest declaring its compatible <code>schema_version</code> and the specific normalized events it subscribes to, ensuring the Dispatcher never routes unsupported payloads.
GitHub API Executor	Executes approved operations against the GitHub API (strictly pinned via <code>X-GitHub-API-Version</code> headers) using installation-scoped tokens derived from private keys stored in a secure vault (e.g., AWS Secrets Manager). Enforces idempotency via known-bot markers and delivery-ID tracking. Handles rate limiting with exponential back-off. Retries transient failures. Records actual mutation results for the audit log. To safely recover from partial-success retries (e.g., label applied but comment failed), the executor verifies the current GitHub state before re-attempting any mutation.

Audit Logger / Store

Records every decision using a structured, append-only logger (e.g., Pino): event context, config version used, selected policy module, intended operations, actual API results, and failure reasons, enforcing a 90-day retention policy. **Storage model:** Phases 1–2 use Pino JSON to stdout, ingested by the hosting platform’s log aggregator. Phase 3 onward appends to a PostgreSQL audit table with a 90-day row TTL enforced by a scheduled purge job. Read access is restricted to the app’s deployment credentials and named maintainers. No separate audit UI is in scope for this mentorship. Enables debugging, regression analysis, and maintainer trust.

3.3. Architectural Principles

1. **One trust boundary, at the listener.** Every event is verified before it enters the internal pipeline. No module receives an unverified payload.
2. **Modules are pure.** Policy modules return decisions; they do not perform I/O. This makes them trivially testable.
3. **Config is policy, not code.** Repositories express intent through typed config fields. They cannot inject behavior through config.
4. **Executor handles all mutations.** No module writes to GitHub directly. This makes idempotency and audit logging centrally enforceable.
5. **Audit is first-class.** Every event produces an audit record, regardless of outcome. An event that produces no mutation still produces an audit entry explaining why.
6. **Shell before modules.** The listener → normalizer → router → dispatcher → executor → audit path must work end-to-end before any policy module is attached.

3.4. Component Interaction Diagram

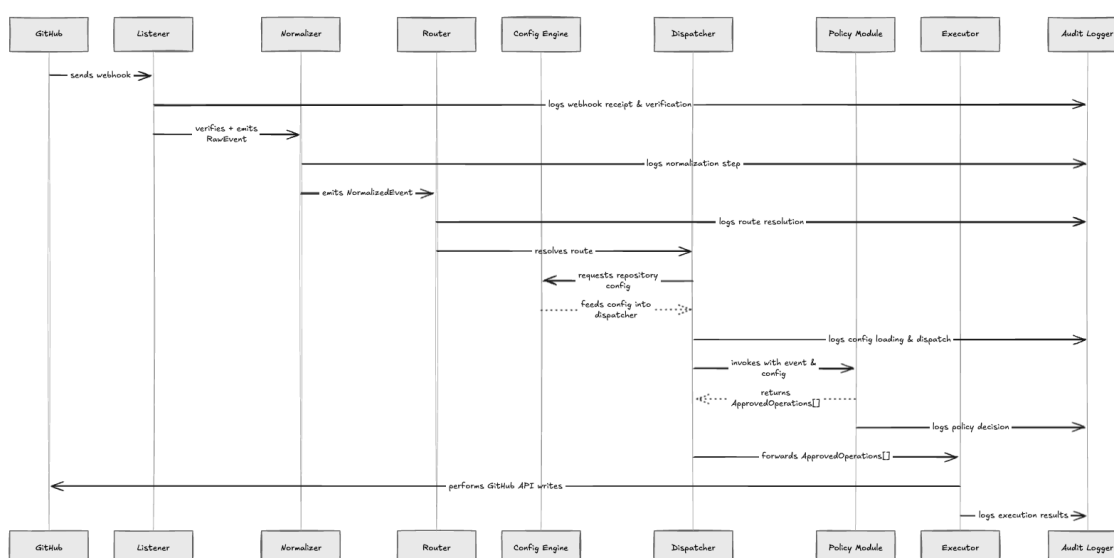


Figure 2: Component Interaction / Data Flow

4. The First Product Slice: /assign

4.1. Why /assign First

`/assign` is the correct first product slice for three reasons:

1. **Forces the full app path.** Handling `/assign` requires: comment parsing, webhook verification, normalisation, routing, config loading, policy evaluation, a GitHub write (assignment), a user-facing comment, and an audit record. No shortcut path is available.
2. **Bounded scope.** Unlike PR quality (many checks) or review-sync (complex state machine), `/assign` is a single command with well-understood guards and a clear success state.
3. **Sophie's explicit direction.** Sophie identified `/assign` as the first candidate. Review-sync is valuable but becomes the second or third module *after* the app shell is proven.

4.2. Sequence Diagram: /assign Full Path

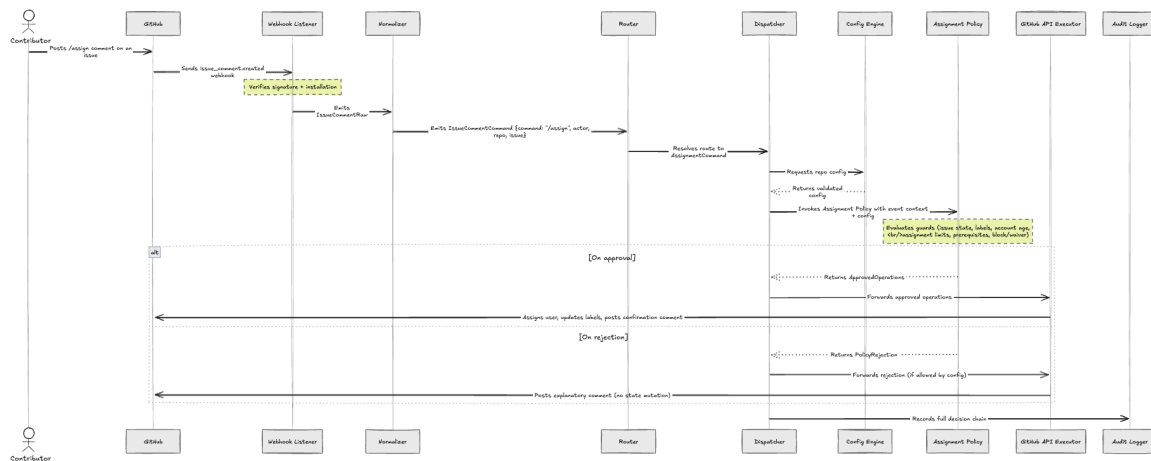


Figure 3: `/assign` Sequence Diagram

4.3. Minimum Behaviour Specification

1. A contributor comments `/assign` on an issue.
2. The listener verifies the webhook signature and installation scope.
3. The normalizer creates an internal `IssueCommentCommand` event.
4. The router detects `/assign` and routes to `AssignmentCommand`.
5. The dispatcher loads repo config and invokes `AssignmentPolicy`.
6. The policy checks:
 - Issue open/closed state
 - Blocking labels (e.g. `in-progress`, `claimed`)
 - Account age check (if enabled in config)
 - Per-user assignment limit (if enabled in config)
 - Prerequisite issues or skill labels (if enabled in config)
 - Maintainer override labels / waiver permissions

7. On success: executor assigns the user, updates labels if configured, posts a confirmation comment, and writes an audit record.
8. On failure: the app writes *no* state mutation except an optional explanatory comment permitted by config. It never silently fails.

4.4. Guard Matrix For `/assign`

Guard	Default	Config override	Failure action
Issue open?	Required	Not configurable	Reject silently
Not already assigned?	Enforced	<code>max_assignments:</code> <code>N</code>	Reject + comment
Account age check	Disabled	<code>min_account_age_days:</code> <code>N</code>	Reject + comment
Prerequisite issues	Disabled	<code>prerequisites:</code> <code>[]</code>	Reject + comment
Blocking labels	<code>in-progress</code>	<code>block_labels:</code> <code>[]</code>	Reject + comment
Waiver permission	None	<code>waiver_team</code>	Skip guard
Maintainer override	By team	<code>maintainer_team</code>	Skip all guards

4.5. Why `/assign` Before Review-Sync

Review-sync is still valuable. It will become the second or third module after the app shell is proven. The reason for deprioritising it is not technical difficulty but architectural discipline: review-sync is a more complex state machine (involving PR review queues, reviewer assignment, and label transitions) that risks polluting the app shell with module-specific concerns before the shell itself is trustworthy.

5. Configuration Model

5.1. Config As A Policy Surface

The configuration file is **not** a place for arbitrary behavior hooks or embedded scripts. It is the app's policy surface: a typed, schema-validated document that lets each repository express its governance intent without forking code.

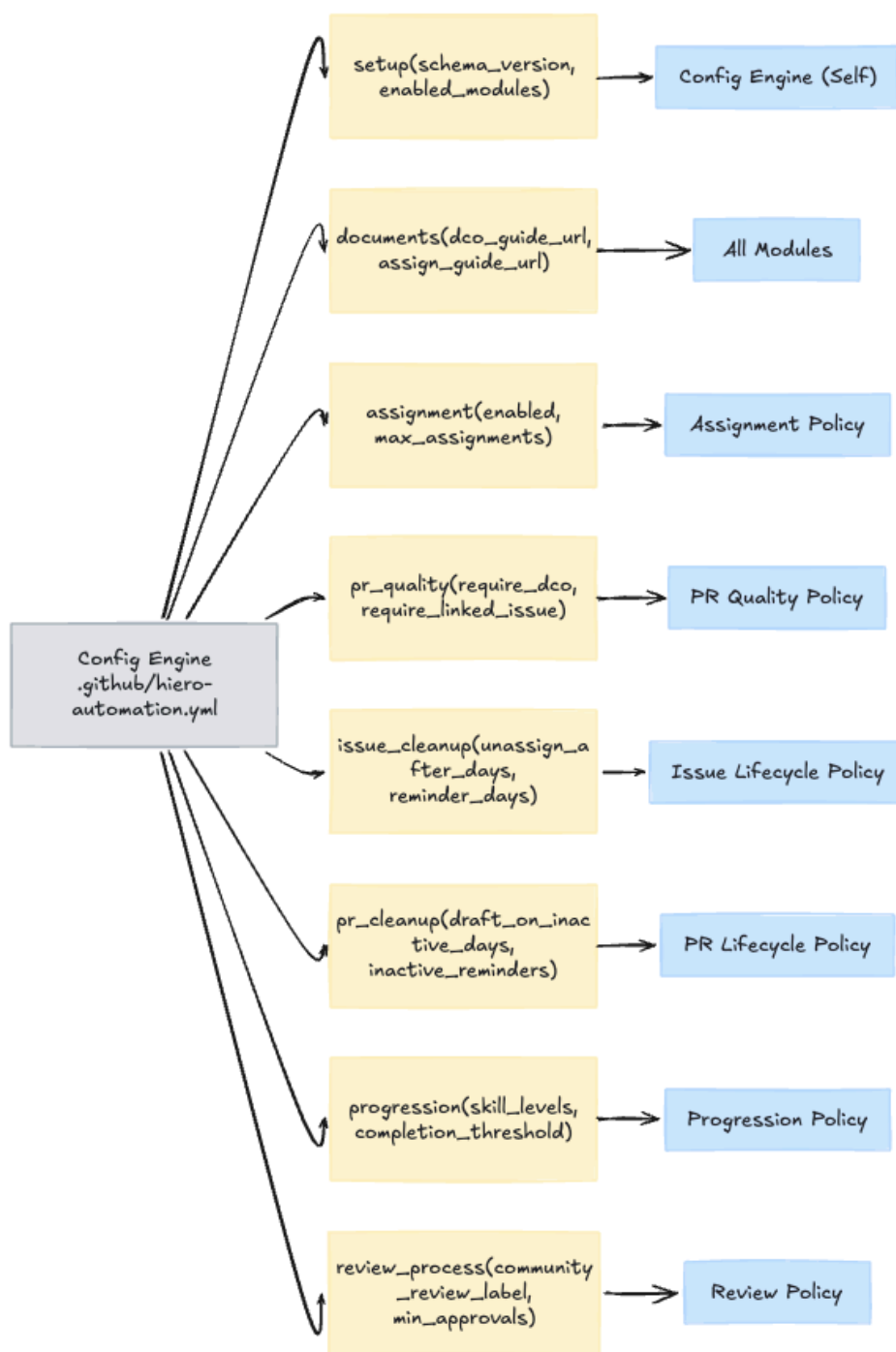


Figure 4: Config Module Map

5.2. Config Sections Reference

Config section	Example fields	Policy module owner
setup	schema_version, enabled_modules[], auto_create_labels	Config Engine

documents	dco_guide_url, gpg_guide_url, assign_guide_url, merge_conflict_guide_url	Config Engine + all modules
assignment	enabled, min_account_age_days, max_assignments, prerequisites[], waiver_team, maintainer_team, block_labels[]	Assignment Policy
pr_quality	enabled, require_dco, require_gpg, require_linked_issue, require_conventional_title, block_merge_conflict, dashboard_label	PR Quality Policy
issue_cleanup	enabled, unassign_after_days, reminder_days, blocked_labels[], waiver_labels[], allow_unassign_command	Issue Lifecycle Policy
pr_cleanup	enabled, draft_on_inactive_days, close_unlinked_issue_pr, inactive_reminder_days	PR Lifecycle Policy
progression	enabled, skill_levels[], completion_threshold, recommendation_labels[], allow_finalize_command	Progression Policy
review_process	enabled, community_review_label, ready_to_merge_label, min_approvals	Review Policy

5.3. Config Validation Rules

- Config expresses policy, not code. No arbitrary scripts or behavior hooks.
- Config is loaded via the GitHub API using the installation token for that repository. It is never loaded from a local file system.
- Unknown fields on high-risk sections (e.g. `assignment`, `pr_quality`) cause the module to fail closed.
- Unknown fields on low-risk sections (e.g. `documents`) produce a warning in the audit log but allow the module to proceed.
- Each module has an `enabled` flag. If `enabled: false`, the module does not load config, does not process events, and does not write state.
- The `schema_version` field is required. Schema upgrades will support a 30-day backward-compatibility window where the previous version is gracefully mapped to the new schema, emitting a deprecation warning in the audit log rather than failing

immediately.

- Missing Config Fallback: If the repository lacks a `.github/hiero-automation.yml` file (404), the Config Engine fails open safely by returning a virtual config where all modules are set to `enabled: false`.
- Repo-specific values (labels, team names, thresholds, doc links) are configurable. Shared behavior (command parsing, label mutation safety, idempotency, audit formatting) is not configurable.

5.4. Sample Config File

```
# .github/hiero-automation.yml
schema_version: "1"

setup:
  enabled_modules:
    - assignment
    - pr_quality

documents:
  assign_guide_url: "https://docs.hiero.example/contributing/assign"
  dco_guide_url: "https://docs.hiero.example/contributing/dco"

assignment:
  enabled: true
  min_account_age_days: 7
  max_assignments: 1
  maintainer_team: "hiero-maintainers"
  waiver_team: "hiero-trusted-contributors"
  block_labels:
    - "in-progress"
    - "claimed"

pr_quality:
  enabled: true
  require_dco: true
  require_linked_issue: true
  require_conventional_title: true
```

6. Boundaries

6.1. Boundary Map

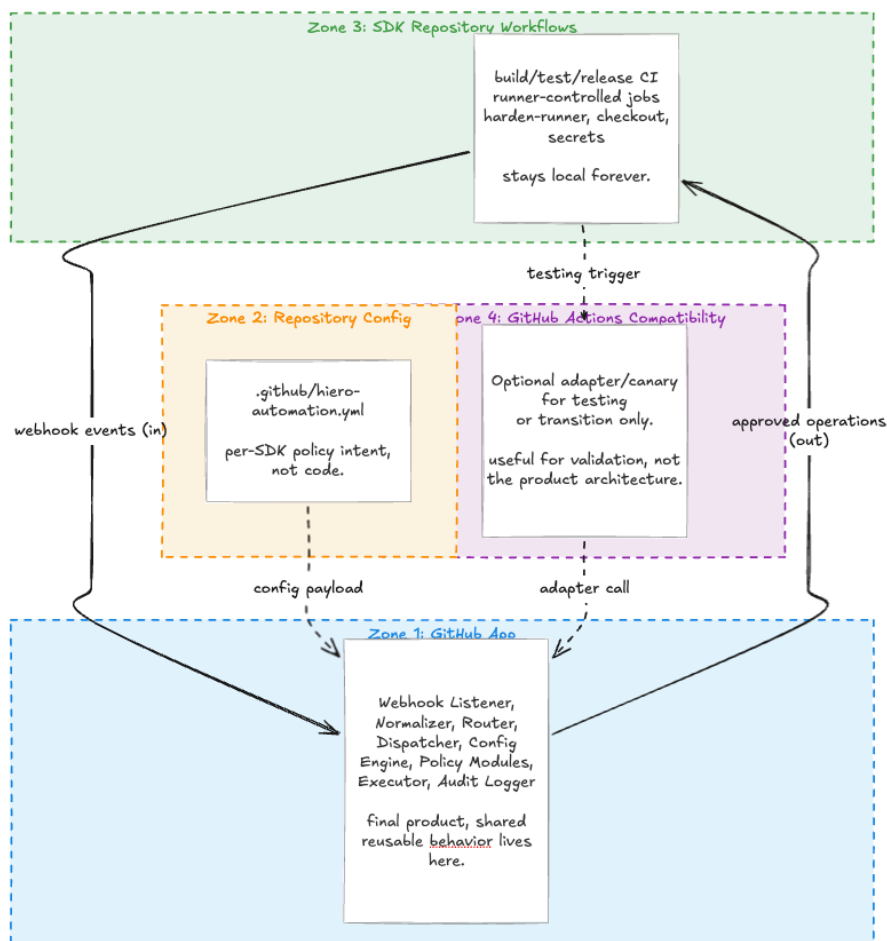


Figure 5: Boundary Diagram

6.2. Boundary Table

Boundary	Stays there	Reason
GitHub App	Webhook listener, event normalizer, router, dispatcher, policy modules, executor, audit logging.	This is the final product and the place where reusable automation behavior belongs.
Repository config	Labels, teams, thresholds, guide URLs, enabled modules, schedules, waiver labels, and policy choices.	SDKs need policy control without forking code.

SDK repository workflows	Build, test, release workflows, repo-specific CI, runner-controlled jobs, artifact handling, harden-runner, checkout, secrets.	Not every workflow is an app workflow. Some should remain local forever. These define the security boundary and event context for each repository.
GitHub Actions compatibility path	Optional adapter/canary for testing or transition.	Useful for validation, but Sophie's feedback says it should not drive the architecture.
Human maintainer control	Approvals, overrides, disabling modules, reviewing rollout, and incident decisions.	Automation should reduce maintenance load, not remove accountability.

6.3. What Does Not Cross The Boundary

- Workflow YAML triggers, permissions, concurrency rules, `if:` guards.
- Harden-runner configuration and runner selection.
- Artifact upload/download wiring.
- Repository secrets and tokens.
- Multi-job orchestration where job boundaries reflect security boundaries.
- Repository-owned config files (the App reads them; it does not own them).

6.4. Legacy Code To Be Retired

The following table inventories the specific legacy artefacts that will be retired as each App module proves parity. No artefact is removed until its replacement module has passed fixture-parity tests and a controlled pilot in at least one repository.

Legacy artefact	Why it must go	Retired after
Per-repo <code>assign.py</code> scripts	Duplicated across SDKs with divergent guard logic, no shared tests, and no audit trail.	Phase 3 (<code>/assign</code> guards proven)
Per-repo <code>review_sync.py</code> scripts	Complex state machines copied between repos with inconsistent label handling and no shared parity baseline.	Phase 5 (review module proven)

GitHub Action wrappers embedding policy logic	Policy decisions buried inside YAML <code>if</code> : guards and inline scripts. Impossible to unit-test in isolation. Invisible to the audit trail.	Per module, after pilot
Scattered API retry and error-handling code	No idempotency keys. No bot-marker checks. No delivery-ID deduplication. Risk of duplicate writes on transient failures.	Phase 1 (executor centralises all writes)
Hardcoded label names and thresholds	Constants embedded directly in scripts instead of validated config. Different values per repo with no schema enforcement or version tracking.	Phase 1 (config engine replaces)
Manual fork-specific canary workflows	Served their purpose as architecture probes and proof-of-concept. No longer needed once the App Shell handles webhooks directly via GitHub App installation.	Phase 2 (sandbox app replaces canary)

Teardown Safety Rule

No legacy script is deleted from an SDK repository until: **(a)** its replacement policy module passes all fixture-parity tests, **(b)** a controlled pilot has run in at least one upstream repository for one week with no unexpected mutations, and **(c)** the responsible maintainer has explicitly approved the removal PR. Premature teardown is as dangerous as premature migration.

7. Security And Reliability Model

7.1. Threat Model

Threat	Attack vector	Mitigation
Forged webhook	Attacker sends a crafted payload to the listener endpoint.	HMAC-SHA256 signature verification before any processing. Reject on mismatch.

Replayed delivery	Attacker replays a valid, previously processed delivery ID.	Delivery ID deduplication window (24h). Idempotency keys per issue/PR.
Config injection	A repo config attempts to expand permissions or inject behavior.	Zod schema validation. Unknown high-risk fields cause module to fail closed.
Comment injection	Malicious /assign target or command with crafted arguments.	Strict command parser. Arguments validated against known patterns.
Duplicate writes	Retries or concurrent deliveries produce double comments or labels.	Idempotency keys and bot-marker checks before any GitHub write.
Module coupling	A growing module starts referencing internals of another module.	Policy modules are pure functions with no module-to-module imports. Dispatcher mediates all cross-module data via the event context.
Over-broad permissions	App receives installation token with write access to unrelated resources.	Minimum necessary GitHub App permissions. Installation-scoped tokens only. Permissions documented per module.

7.2. Reliability Safeguards

- Verify webhook signatures and delivery IDs before processing.
- Use GitHub App installation-scoped permissions, not personal access tokens.
- Normalize events and validate route eligibility before dispatching.
- Validate config schema; reject unknown high-risk fields.
- Idempotency keys, known-bot markers, and per-issue/PR concurrency locks.
- All decisions auditable: input, config version, policy module, intended mutation, actual mutation, and failure reason.
- GitHub Actions adapter treated as a testing/compatibility path with visible hardening and pinned references; never as the production architecture.
- Rate-limit handling with exponential back-off in the executor. No module is responsible for retry logic.
- Lifecycle & Abuse Defenses: The listener drops payloads exceeding standard size limits to prevent DoS, and processes `installation.deleted` webhooks to gracefully purge orphaned tokens and cached state.

Webhook delivery deduplication (operational note)

Operational contract. To avoid duplicate side-effects from retried or replayed webhook deliveries (double-assignment, duplicate comments, conflicting label changes), the system requires a durable delivery deduplication store and an atomic claim pattern. The note below records recommended implementation choices and operational checks; it is intentionally concise and non-normative.

Recommended backing stores:

- **Redis:** use SET key value NX PX (ms) to create a deduplication key with a TTL.
- **Postgres:** use a `delivery_log` table with a unique constraint on (`app_id`, `delivery_id`) and an `expires_at` timestamp.

Key schema and metadata. Use keys such as `app:<app_id>:delivery:<delivery_id>`. Stored metadata SHOULD include repository id, event type, received timestamp, and a short processing trace id.

Atomic claim pattern. On receipt, after signature verification and basic validation, attempt an atomic create/insert for the dedup key:

- If the create succeeds, proceed with normal processing.
- If the key already exists, treat the delivery as a duplicate and skip further non-idempotent actions.

Failure mode (recommended). If the dedup store is unreachable, the recommended default is *fail-closed*: do not perform non-idempotent changes; instead enqueue the event for manual reconciliation and emit an alert.

Observability. Expose metrics such as `dedup.store.calls`, `dedup.store.errors`, and `dedup.duplicate.count`, and record an audit entry mapping each `delivery_id` to the processing decision.

Installation token lifecycle (operational note)

Problem. GitHub App installation tokens expire after one hour. The executor and long-running paths must not depend on a token fetched at start and left to expire mid-execution.

Recommendation. Implement a token cache layer per installation with the following behaviour:

- Cache entry TTL: 55 minutes (i.e., a 5-minute safety buffer).
- On access: if the remaining token TTL is under 5 minutes, synchronously refresh before use.
- Refresh operation: single-flight per installation (lock or in-process mutex) to avoid stampeding refreshes under load.
- Persistent options: in-process cache is acceptable for single-worker deployments; use Redis or a shared cache for multi-worker clusters.
- Proactive refresh: background refresher may renew tokens earlier to amortize latency outside request critical path.

Failure handling and circuit breaker. If token refresh calls to GitHub fail repeatedly, open a circuit breaker for the installation and redirect events to a durable queue for manual/operator reconciliation until health is restored. Emit alerts when the circuit trips and expose metrics: `token.refresh.calls`, `token.refresh.errors`, `token.circuit.open`.

Audit and observability. Log token refresh attempts and reasons for failure, and ensure API calls that fail with 401 include the token refresh trace id in the audit record to make post-mortem triage straightforward.

Config loading cache and stampede protection (operational note)

Problem. Loading `.github/hiero-automation.yml` from the GitHub API for every event does not scale and can exhaust rate-limit budget under concurrent activity.

Recommendation. Add a configuration cache and miss-coordination mechanism with the following behaviour:

- Cache scope: per repository (and default branch or SHA context, if relevant to policy behaviour).
- Positive cache TTL: 5 minutes (tunable by deployment).
- Stampede lock: on cache miss, allow exactly one fetch per repository; concurrent requests wait for that fetch result instead of fan-out.
- Negative cache: if config is absent, cache the "not found" result for a short TTL (for example 60 to 120 seconds) to prevent repetitive misses.
- Invalidation: on config-file change events, invalidate the repository cache key immediately.

Failure handling. If config fetch fails due to transient upstream errors, continue with last-known-good config when available; otherwise apply a documented safe default policy and emit an alert.

Observability. Track and alert on cache behaviour using metrics such as `config.cache.hit`, `config.cache.miss`, `config.cache.negative.hit`, `config.fetch.calls`, and `config.fetch.errors`. Audit records should include cache decision and resolved config version.

Operational observability and alerting specification

Objective. The app must be operationally observable in production, not just auditable after incidents. Audit logs and runtime telemetry are complementary and both are required.

Metric counters and timings. At minimum, expose and dashboard:

- `events.received.count` (per event type and repository)
- `events.processed.count` (successful end-to-end handling)
- `events.failed.count` (failed handling, by error class)
- `github.api.write.latency.ms` (p50, p95, p99)
- `github.api.write.error.count` (by status class: 4xx, 5xx)

Structured log format. Emit JSON logs for each pipeline stage with stable fields: `timestamp`, `level`, `event_id`, `delivery_id`, `installation_id`, `repository`, `route`, `module`, `decision`, `error_class`, `error_message`, and `correlation_id`. The `correlation_id` must be propagated from listener to executor and included in audit records, metric labels where appropriate, and external alerts.

Alert policy. Define actionable alerts with on-call ownership:

- Error-rate alert: `events.failed.count / events.received.count > 5%` for 5 consecutive minutes.
- API degradation alert: sustained increase in `github.api.write.latency.ms` p95 or elevated 5xx write errors.
- Queue/backlog alert (if queue is enabled): lag above operational threshold for longer than the configured drain window.

Health endpoints for orchestrators.

- `/healthz` (liveness): process is running and event loop is responsive.
- `/readyz` (readiness): dependencies available (GitHub API token path, cache, queue, and config fetch path).

If readiness fails, the instance should be removed from serving traffic while remaining alive for diagnostics.

Runbook linkage. Each alert must link to a short runbook section with: initial triage steps, known failure modes, rollback/disable instructions per module, and escalation contacts.

Hosting and deployment model gate (operational note)

Decision required before Phase 1. The runtime model must be selected before implementation of the app shell begins. At minimum, choose between:

- containerised Node.js/TypeScript long-running service (e.g., Cloud Run or Railway)
- stateless function runtime (serverless invocation model)

Why this is gating. The hosting model determines core stateful component design: token cache behavior, delivery deduplication store, configuration cache strategy, concurrency control, and retry/backoff behaviour.

Minimum architecture commitments.

- If long-running service is chosen, define worker concurrency model, process-level cache limits, and restart semantics.
- If stateless function runtime is chosen, require externalized state for token cache, deduplication, and config cache from day one.
- For either model, define network exposure, secret delivery method, egress controls, and rollout strategy (blue-green or equivalent).

Required artifact. Record the decision as a short ADR before Phase 1 starts, including selected runtime, dependency assumptions, and explicit impact on `token.refresh`, `dedup.store`, and `config.cache` implementations.

Operational consequence if deferred. Deferring this choice risks building a Phase 1 shell that is structurally incompatible with production hosting constraints and forces rework of foundational reliability components.

Hosting Decision — Resolved

Selected runtime: Long-running Node.js/TypeScript container deployed to **Railway** (free tier for development, scalable for production).

Supporting services:

- **Redis** for delivery deduplication, entity-level concurrency locks, installation token cache, and configuration cache.
- **PostgreSQL** for persistent audit log storage (Phase 3 onward).

Rationale: A long-running container is preferred over serverless for this project because: (a) GitHub App installation tokens (1-hour TTL) benefit from a persistent in-process token cache; (b) delivery deduplication requires durable state; (c) serverless cold-start latency is problematic for webhook response SLAs (GitHub expects <10s); and (d) a single-container model is simpler to debug during mentorship development.

Webhook tunnel for local development: During Phase 1, use **smee.io** (GitHub-maintained) as the webhook proxy for localhost development. The smee channel URL is registered as the webhook URL during development and swapped for the Railway production URL at Phase 2 sandbox deployment.

Concurrency model and lock semantics (operational note)

Problem statement. Concurrent webhooks can target the same issue or PR within milliseconds. Guard logic alone is insufficient without a concrete serialization primitive.

Lock scope (required). Use per-entity lock keys to serialize state-changing operations:

- Issue key: `lock:repo:<repo_id>:issue:<issue_number>`
- PR key: `lock:repo:<repo_id>:pr:<pr_number>`

Allowed lock primitives.

- Redis: SET key value NX PX (ttl_ms) with ownership token and compare-and-delete unlock.
- Postgres advisory lock: transaction-scoped advisory lock derived from repository and entity identifiers.

Acquisition and release contract.

- Acquire lock before policy evaluation that can lead to mutation.
- Keep lock TTL bounded (for example 10 to 30 seconds) and renew only if execution is still active.
- Release lock on completion and on all error paths.
- If lock owner is lost or expired, treat in-flight operation as unknown outcome and require idempotent re-check before retrying writes.

Lock acquisition failure behaviour (must be explicit). Default policy:

- Retry briefly with jittered backoff.
- If still unavailable, enqueue for delayed retry.
- If queueing is unavailable, fail safely with a non-mutating response and optional explanatory comment (config-controlled).

Observability. Expose `lock.acquire.success`, `lock.acquire.failure`, `lock.wait.ms`, and `lock.timeout.count`. Include lock key and correlation id in audit records for race-condition triage.

8. Testing Strategy

Testing belongs *after* architecture. It proves the app works; it is not the main architectural story. Every phase has its own testing gate. A phase is not complete until its gate passes.

8.1. Testing Layers

Layer	Purpose	First required phase
Unit tests	Policy decisions, command parsing, config validation, guard evaluation, and idempotency behavior. Each policy module tested in isolation as a pure function.	Phase 1 (app shell)

Fixture / golden tests	Real Python and C++ examples converted into golden cases. The shared implementation must reproduce the expected outcome for every fixture before it handles live events.	Phase 2 (/assign minimal)
Integration tests	Mock GitHub API interactions for comments, labels, assignments, and failure handling. Tests the executor and audit logger together.	Phase 2
Schema / config tests	Config loading, validation, default application, and rejection of invalid or unknown high-risk fields.	Phase 1
Security / negative tests	Invalid signatures, replayed delivery IDs, malformed commands, config injection attempts. All must be rejected cleanly.	Phase 1
Sandbox app test	Private/fork repositories installed with the GitHub App to verify the full webhook flow end-to-end.	Phase 2
Dry-run / canary	End-to-end validation after the app path exists. Confirms intended behavior before writes are enabled. Not the product architecture itself.	Phase 3
Production pilot	One repo, one module, named owner, rollback plan, documented observation window.	Phase 3

8.2. Coverage Target

- 90%+ branch coverage on all state-mutating logic (executor, policy modules).
- 100% fixture coverage: every accepted expected behavior from Python/C++ must have a corresponding fixture test in the central suite before that behavior is shipped.
- Security tests must run on every CI push. A failed security test blocks merge.

8.3. Testing Gates By Phase

1. **Phase 1 closes** when: config/schema tests pass, security tests pass, and synthetic events route through the full pipeline with correct audit output.
2. **Phase 2 closes** when: **/assign** unit tests pass, fixture parity is established, and a sandbox end-to-end run completes without duplicate writes.
3. **Phase 3 closes** when: all guards have fixture-backed tests and a dry-run pilot completes with no unexpected mutations.
4. **Phase 4 closes** when: PR quality module has independent fixture coverage and does not couple to assignment module internals.

5. **Phase 5 closes** when: Issue Lifecycle and PR Lifecycle modules each have config sections, unit tests, fixture coverage, audit log output, and independent **enabled** flags. Modules do not reference each other's internals.

9. Phased Delivery Strategy

9.1. How To Read This Roadmap

- **Phases** describe what truth must be established and what architecture must exist before moving to the next phase.
- **Milestones** are the calendar checkpoints where mentor and maintainers review evidence and decide whether to proceed.
- **Exit criteria** are the specific, observable conditions that close a phase. A phase is not complete until its criteria are met, regardless of calendar date.
- **Rollback plans** are written before work begins, not after a problem occurs.

9.2. Migration Strategy: Vertical Feature Slicing

The migration does not proceed repository-by-repository (“migrate all of Python, then all of C++”). It proceeds **feature-by-feature across all repositories simultaneously**:

1. **Pick one automation feature** (e.g. `/assign`).
2. **Build the policy module** as a pure decision function inside the App Shell.
3. **Prove parity** with fixture-backed tests using golden cases extracted from both Python and C++ existing behaviour.
4. **Pilot in one repository** (sandbox fork first, then one upstream repo with explicit maintainer approval).
5. **Retire the legacy script** in that repository only after the pilot passes a one-week observation window with no unexpected mutations.
6. **Expand to the next repository** by adding a `.github/hiero-automation.yml` config file and setting `enabled: true` for the module.
7. **Move to the next feature** only after the current feature is stable across all participating repositories.

This vertical strategy has three advantages over horizontal (repo-by-repo) migration:

- **Proves the App Shell early.** The first vertical slice (`/assign`) forces the entire eight-stage pipeline to work end-to-end. Every subsequent module reuses the same proven shell without re-verifying the listener, router, or dispatcher.
- **Limits blast radius.** A bug in one policy module affects only that automation feature, not all automations in a repository. Independent **enabled** flags per module enforce this isolation.
- **Enables incremental rollback.** Disabling one module via its config flag does not affect any other module. Rolling back `/assign` does not touch PR quality checks or lifecycle automations.

9.3. Phase Roadmap Diagram

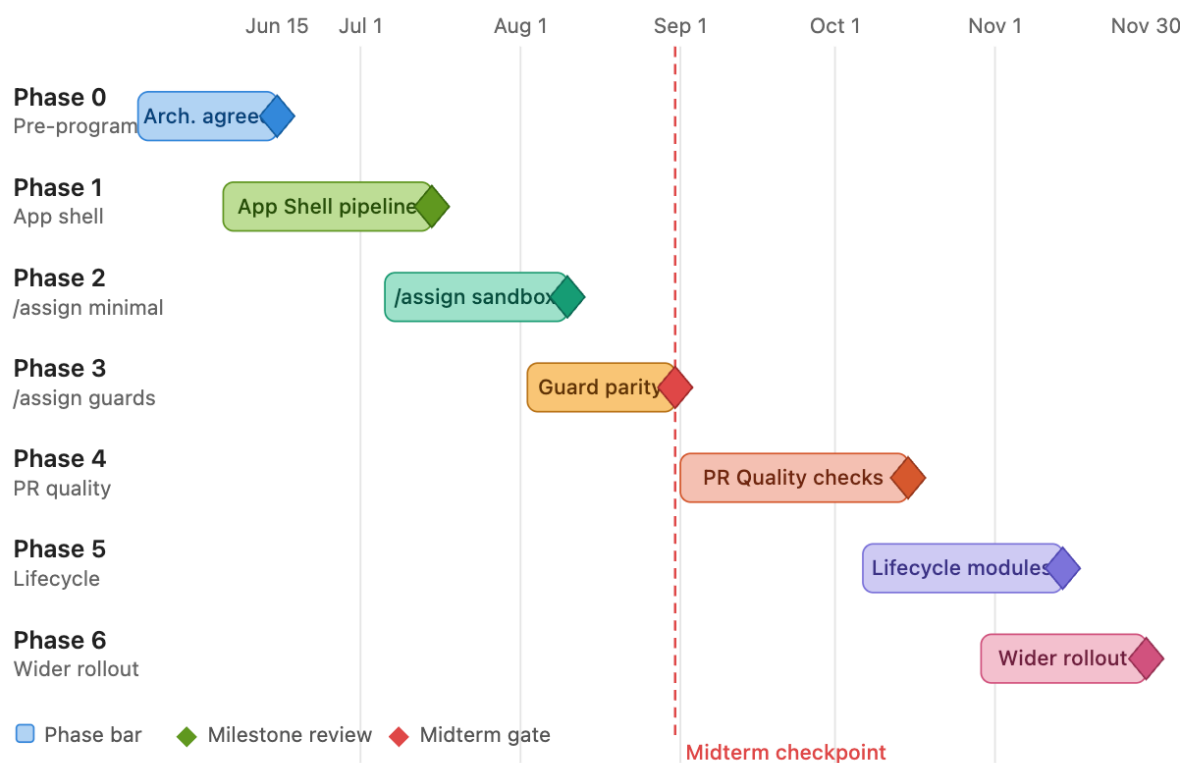


Figure 6: Phase Roadmap / Timeline

9.4. Integrated Phase Table

Phase	Timebox	Build	Exit criteria
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Phase 0 Architec- ture agreement	Pre Jun 15	Agree on the eight-component pipeline. Align on /assign as first slice. Confirm config model direction with mentor. Register the GitHub App in the Hiero organisation with minimum permissions (Issues write, Pull Requests read/write, Contents read). Generate private key and store in secrets vault. Set up smee.io channel for local webhook development. Select hosting platform (Railway).	Mentor and maintainer agree that this is the product architecture and the correct starting slice. GitHub App is registered with correct permissions. Private key is stored securely. Smee.io channel is active. Hosting platform decision is recorded as a short ADR. No upstream SDK depends on candidate repos yet.
Phase 1 App shell	Jun 15 – Jul 15	Webhook listener with signature verification. Event normalizer. Router with declarative route registration. Dispatcher with module registry. Config engine with schema validation. Audit logger. Smee.io webhook tunnel for local development. GitHub Actions CI/CD workflow for the app's own deployment (merge to main triggers deploy to Railway staging). <i>No policy modules yet.</i>	Synthetic webhook events route correctly through all pipeline stages. Audit records are produced. Security tests pass. Config schema validation tests pass. App deploys via GitHub Actions workflow to staging. Rollback is achieved by reverting the merge.
Phase 2 /assign minimal	Jul 16 – Aug 10	Assignment Policy: issue eligibility, assignment write, explanatory comment, audit event. No guards yet (guards come in Phase 3). Sandbox end-to-end test.	Works in sandbox. No duplicate writes. Failure paths produce clear audit records. Fixture parity established against at least three Python and three C++ golden cases.

Phase 3 /assign guards	Aug 11 – Aug 31	Account age check, max assignments limit, prerequisites check, waiver labels, maintainer override, block labels. Dry-run pilot in one repository.	All guard conditions have fixture-backed tests. Dry-run output reviewed by at least one maintainer. No unexpected mutations during dry-run pilot. Midterm checkpoint by Aug 31.
Phase 4 PR quality module	Sep 1 – Oct 15	DCO check, GPG check, merge conflict detection, linked issue requirement, conventional title check, dashboard label. Implemented as a separate module with no coupling to Assignment Policy internals.	Runs as an independent module. Has its own config section, fixture suite, and independent enabled flag. Does not share internal state with Assignment Policy.
Phase 5 Lifecycle modules	Oct 16 – Nov 14	Issue Lifecycle Policy (/unassign , /blocked , stale reminders, auto-unassign). PR Lifecycle Policy (draft conversion, inactive reminders). Two modules, each with approximately 15 days of dedicated development. Progression Policy and Review Policy are designed and architecturally integrated but production-enabled post-mentorship (see Section 12).	Each module has its own config section, unit test suite, fixture coverage, audit log output, and independent enabled flag. Modules do not reference each other's internals. Fixture-parity tests pass for both modules.

Phase 6 Wider rollout	Nov 15 – Nov 30	Add repositories only after module stability and explicit maintainer approval per repo. Adoption checklist, rollback path, and community issues per repo.	At least one named non-sandbox repository has adopted the app with a named module owner, documented rollback plan, and passed a one-week observation window. Org-wide rollout to all Hiero SDKs follows post-mentorship, as each repository's maintainers approve adoption.
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9.5. Midterm Checkpoint (End Of August)

The midterm checkpoint is a formal review at the end of Phase 3. Mentor and maintainers review whether the app shell is ready for controlled writes.

Midterm Decision Gate

Required evidence at midterm:

- Live demo of the `/assign` path through the full pipeline.
- Dry-run output reviewed and understood by at least one Python maintainer.
- Fixture parity established for `/assign` golden cases.
- Rollback plan documented and tested.
- At least one maintainer can operate the audit log without assistance.

Decision: Proceed to Phase 4 (PR quality), narrow scope, or halt.

9.6. Acceleration Path

If Phases 1–3 land cleanly ahead of schedule, the late-program window should be used to start one bounded stretch path: either a minimal GitHub App adapter spike for future Probot hosting, or a tightly scoped second-repo canary. The rule is: *accelerate only by adding bounded next-step proofs; never by cutting testing, parity, or security gates.*

9.7. Rollback Plans By Phase

Phase	Rollback plan
Phase 1	App shell has no production effects. Rollback means stopping the app service. No SDK repository is affected.
Phase 2	Sandbox only. Rollback means removing the test installation from the fork. No upstream repo is affected.
Phase 3	Disable the dry-run workflow or set <code>dry_run: true</code> . No production state changed during dry-run. If a write pilot was enabled, set <code>assignment.enabled: false</code> in the repo config.

Phase 4	Set <code>pr_quality.enabled: false</code> per repo. The Assignment module continues independently.
Phases 5, 6	Disabling a module in repo config takes effect once the 5-minute config cache expires. For immediate, panic-level mitigation across all repos, the hosting container exposes a global <code>DISABLE_ALL_WRITES</code> environment variable kill-switch to bypass the cache instantly.

10. Community Issue Strategy

10.1. Opening Principles

Community issues should be opened around safe, reviewable units that build the app architecture without forcing risky production behavior changes.

1. Open issues only after the architecture or policy boundary for that phase is decided.
2. Start with documentation, schema, fixtures, checklists, and behavior analysis before opening privileged workflow or production behavior changes.
3. Use community issues to widen already-understood work, not to discover architecture live in production-sensitive repositories.
4. Keep contributor-facing assignment behavior and privileged workflow rewiring maintainer-owned until the shared model is proven in smaller steps.
5. Every community issue must have: a clear scope, a definition of done, and a label indicating its safe-to-open phase.

10.2. Issue Label System

Label	Open from	Meaning
<code>good-first-issue</code>	Phase 1	Safe for any community contributor. No production risk.
<code>architecture-doc</code>	Phase 0	Writing or reviewing ADRs, architecture docs, decision records.
<code>fixture</code>	Phase 1	Collecting, converting, or reviewing golden test fixtures.
<code>schema</code>	Phase 1	Working on the config schema, docs, validation rules, or defaults.
<code>policy-module</code>	Phase 2	Implementing or testing a pure policy decision function.
<code>security-review</code>	Phase 1	Reviewing threat model, idempotency, permissions, or webhook verification.
<code>rollout-doc</code>	Phase 3	Adoption checklists, per-SDK rollout guides, onboarding walkthroughs.

maintainer-only	Never open	Production behavior changes, privileged config, C++ assignment before mapping.
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10.3. Suggested Issue Backlog By Phase

Phase	Safe to open	Keep maintainer-owned
Phase 0	Architecture ADRs for listener, router, dispatcher. Config engine design doc. Component responsibility table review.	Final architecture decisions. Module boundary decisions.
Phase 1	Config schema sections. Schema compatibility and versioning contract. Audit log format spec. Security checklist review. Unit test scaffolding.	Official ownership, branch protection, release control.
Phase 2	/assign command parser tests. Issue eligibility fixture collection. Golden case conversion from Python/C++ existing behavior.	Upstream /assign production change. Maintainer override logic.
Phase 3	Guard fixture tests (account age, max assignments, prerequisites). Dry-run review checklist. Adoption checklist polishing.	Write-mode pilot. Guard behavior decisions.
Phase 4	PR quality fixture collection. DCO check unit tests. Conventional title parser tests. Linked issue validator tests.	PR quality write pilot. Config schema extensions.
Phase 5	Lifecycle module fixture conversion. Progression skill level spec. Review process label inventory. Simple before/after docs.	Production lifecycle module behavior. C++ contributor-facing changes.
Phase 6	Per-SDK adoption checklist. Demo and onboarding docs. Sandbox walkthrough guide.	Per-repo production rollout decision.

10.4. Suggested First Community Issue Batch

The safest first batch combines Phase 0/1 documentation work with Phase 2 fixture work. These are high-leverage areas where contributors create momentum without quietly changing behavior that maintainers are responsible for defending.

1. Write the ADR for the listener → normalizer → router → dispatcher design decision.
2. Write the ADR for the Config Engine and module policy surface.

3. Define the config schema sections with types, defaults, and documentation for **setup** and **assignment**.
4. Collect Python/C++ golden cases for **/assign** decisions (assign success, account age fail, already assigned, blocking label).
5. Write the audit log format specification.
6. Build the SDK caller adoption checklist for maintainers.
7. Write the security review checklist: webhook verification, config trust, idempotency.
8. Create a sandbox repository walkthrough for **/assign** and audit log output.

10.5. Issues To Delay Until Later

- Full C++ workflow migration issues before Phase 4 mapping work is complete.
- Any issue that changes contributor-facing assignment behavior before Python/C++ behavior matrix is agreed.
- Production GitHub App hosting or Probot issues before the app shell is stable through Phase 3.
- Wrapper issues whose main value is reducing YAML while hiding harden-runner, permissions, or trust boundaries.

11. System Planning And Decision Framework

11.1. Explicit Dependencies

These decisions are open dependencies, not assumptions. The migration should not pretend they are already settled.

Resolved: App Repository Governance

Decision: The GitHub App source code will be hosted in a dedicated, org-owned repository under the Hiero GitHub organisation (e.g., `hiero-ledger/hiero-workflow-app`). The existing `sdk-automations` fork by Darshit serves as the proof-of-concept and development sandbox; the canonical repository is transferred to the Hiero org before Phase 1 begins.

Governance:

- **CODEOWNERS:** The mentee (primary) and Sophie (mentor/reviewer). Post-mentorship ownership transitions to the Hiero maintainer team.
- **Releases:** Approved by at least one maintainer via signed tag.
- **Branch protection:** `main` requires passing CI and one approving review. Force push disabled.
- **Incidents:** Owned by whoever holds the deployment credentials (mentee during mentorship, Hiero ops team post-mentorship).

11.2. What This Plan Demonstrates About System Thinking

Sophie asked whether the planning goes beyond technical implementation. The following elements of this document address that question directly:

1. **Product-first framing.** The architecture is described as a product (a GitHub App with a clear event pipeline), not as a collection of scripts. The mentorship

output should look like a small, robust platform.

2. **Phased, gated delivery.** Phases are not arbitrary time blocks. Each phase has an exit criterion that must be satisfied before the next phase begins. This prevents the common failure mode of shipping half-finished features.
3. **Explicit rollback plans.** Every phase documents its rollback plan before work begins. This demonstrates that production safety is designed in, not bolted on.
4. **Community leverage at the right time.** The issue backlog strategy ensures that community contributors accelerate the project without accidentally touching production security boundaries. This requires thinking about information flow and trust levels, not just code.
5. **Config as a governance contract.** The config model is not a convenience feature. It is the mechanism by which 8+ SDK repositories can each express their own governance policy without forking code. This is a system design decision, not a technical one.
6. **Audit trail as a trust mechanism.** The audit logger is a first-class component because maintainer trust is earned through transparency, not promised through documentation. Every decision the app makes is visible and inspectable.
7. **Boundary discipline.** The boundary table (Section 6) shows that not all automation belongs in the App. CI jobs, artifact workflows, and runner-controlled tasks stay local forever. Over-centralisation is a risk, not just under-centralisation.

12. Post-Mentorship Roadmap

The following modules are fully designed and architecturally integrated in this document but are **not** committed to the six-month mentorship delivery plan. They are production-enabled post-mentorship.

12.1. Deferred Modules

Module	Scope	Status at mentorship end
Progression Policy	/finalize command, skill-level tracking, completion thresholds, recommendation labels.	Config schema defined. Module interface designed. No implementation or fixture suite.
Review Policy	Community review label flow, ready-to-merge label, minimum approvals enforcement.	Config schema defined. Module interface designed. No implementation or fixture suite.

12.2. Post-Mentorship Priorities

1. **Progression Policy implementation.** Build, test, and pilot following the same vertical-slice pattern established during the mentorship (fixture parity → sandbox → dry-run → pilot).
2. **Review Policy implementation.** Same pattern. Review Policy has the most complex state machine and should not be rushed.
3. **Org-wide rollout.** Expand App installation to all remaining Hiero SDK repositories (Java, Go, Swift, Rust, and others) as each repository's maintainers approve adoption.
4. **Operational maturity.** Promote audit storage from PostgreSQL to a dedicated log analytics platform if volume warrants it. Evaluate Probot as a long-term framework migration target.

Honest Scoping

Deferring Progression Policy and Review Policy to post-mentorship is not a failure. It is the direct consequence of the same architectural discipline that makes Phases 1–4 trustworthy: one module at a time, done right, with full fixture parity and controlled pilots. Rushing four modules in 29 days would produce half-finished skeletons that damage the trust this plan is designed to build.

13. Final Recommendation

Final Recommendation

Proceed with the Hiero Workflow App, but anchor it around the final GitHub App architecture from day one. The mentorship output should not look like a collection of workflow migrations. It should look like a small, robust product platform.

The next concrete step is Phase 1: build the webhook listener, event normalizer, router, dispatcher, config engine, and audit logger. No policy modules yet. Every synthetic event should route correctly and produce an audit record before a single policy module is attached.

The first policy module is /assign. It is small enough to build correctly, large enough to prove the full app path, and explicitly identified by Sophie as the right starting point.

PR quality and lifecycle modules come after the app shell is stable. They are not afterthoughts; they are planned modules with their own config sections, fixture suites, and independent enablement flags. But they should not be built before the shell is proven.

The community issue backlog should be opened in phases. Architecture docs, schema work, and fixture collection can begin immediately. Production behavior changes require maintainer approval at every step.

This path gives the project a working, trustworthy proof without sacrificing security visibility, local repository control, or maintainer confidence. Speed is not the goal. Correctness is.